TEAM ACADEMIC RESEARCH | JANUARY - MAY 2024

Optimising one-thumb mobile keyboard layouts

Salman implemented the Python genetic-algorithm pipeline; results are from simulation.

36.5%

Improvement in combined score

26%

Reduction in travel distance

1,000,000

Candidate layouts in reported run

Generate

Candidate layouts

→

Evaluate

Travel + comfort score

→

Select

Higher-scoring parents

→

Vary

Produce next generation

What the figures mean

The report compares evaluated layouts with QWERTY in a simulated one-thumb setting. These figures do not establish an equivalent improvement in human typing speed or clinical accessibility outcomes.

Keyboard Optimization project report, Results | github.com/SyndicateGit/Mobile_Keyboard_Optimization_Genetic_Algorithm

The simulation results and evolved layout

Original report page: convergence plots, the resulting keyboard layout, and the reported comparison with QWERTY.

Results

We simulated 1000 generations with 1000 keyboards per generation. This time 100 top-performing parents are selected to produce 800 children and 100 random keyboards are contributing to each generation. 1,000,000 keyboards were generated for this run.

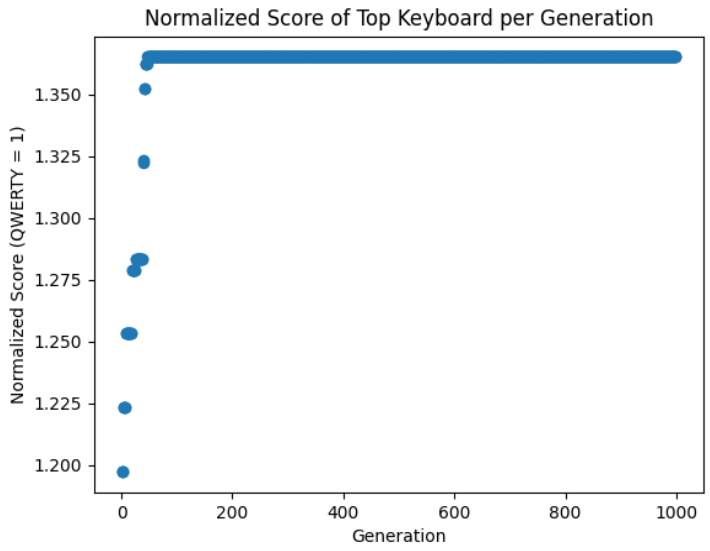


Figure 7 – Normalized score

Figure: 7) Performance Score of top keyboards per generation relative to QWERTY in 1000 generations of 1000 keyboards per generation.

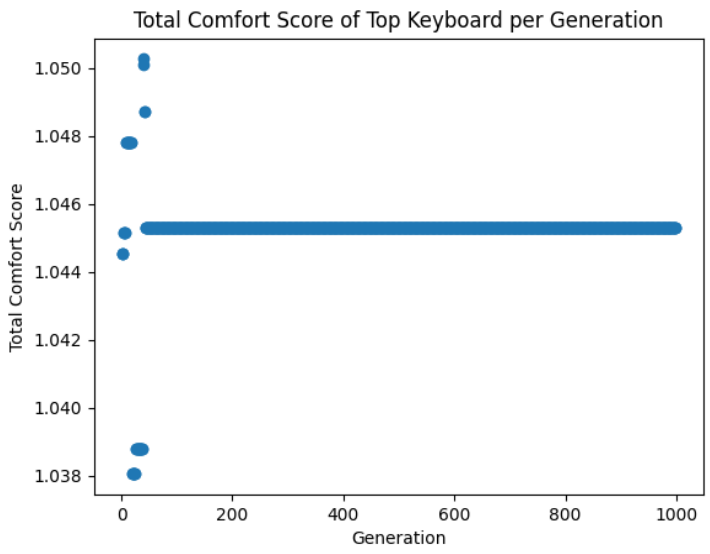


Figure 8 – Comfort score

Figure 08) Total Comfort Score of top keyboards per generation relative to QWERTY in 1000 generations of 1000 keyboards per generation.

We see a compromise of comfort score in favor of reduced total distance traveled by the sudden decrease and plateau at a lower level than the peak.

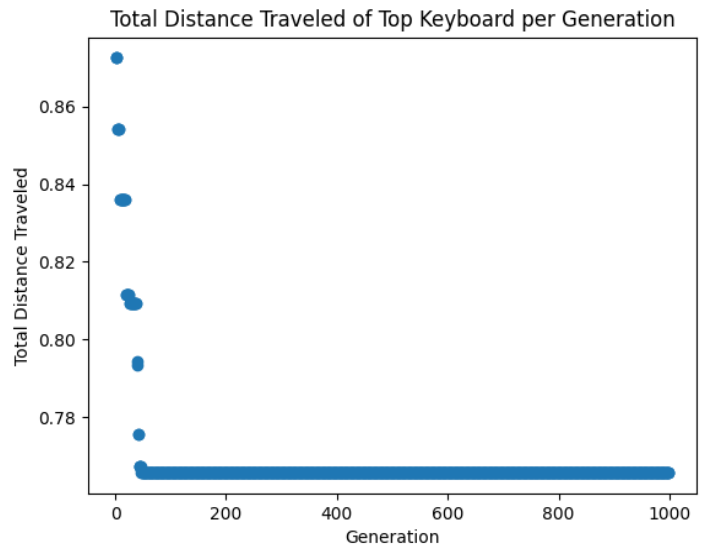


Figure 9 – Total Distance Travelled

Figure: 9) Total distance traveled of top keyboard per generation relative to QWERTY in 1000 generations of 1000 keyboards.

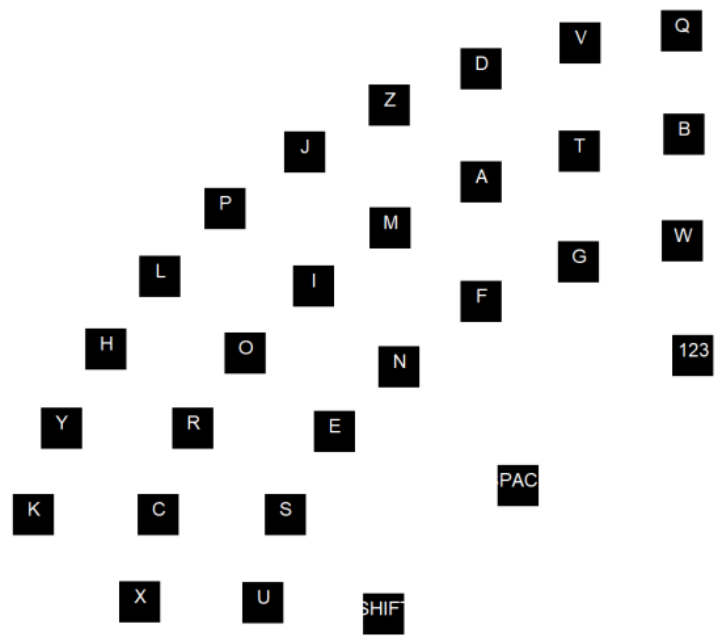


Figure 10 – Keyboard Layout

Figure 10) Example optimized keyboard after 1000 generations.

The top performing keyboard of 1,000,000 keyboard simulation outperforms QWERTY by 36.5% in performance score, 4.5% in comfort, and 26% reduced total distance traveled. Most of the optimizations happened in the first 100 generations before plateauing off. We have noticed that the letter ‘E’ seems to always be near the space and shift key (10/10 of the 50 generation runs). This made sense since it is the